

MOSAIC: auditable multimodal single-cell annotation across donors and incomplete protein panels

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Abstract

Motivation: Fine-grained CITE-seq annotation must remain inspectable under donor shift, incomplete antibody panels and labels absent from training.

Results: We present MOSAIC, a branch-supervised RNA–ADT classifier with margin-based branch arbitration and bounded hierarchy-aware sibling refinement (HSR). In nested leave-one-donor-out evaluation of 161,764 PBMCs and 58 labels, MOSAIC improved weighted and macro F1 over matched early fusion by 0.0040 and 0.0148, respectively, but not accuracy significantly. Direct ablations showed that HSR, knowledge distillation and learned gating did not improve aggregate performance. With 50% proteins masked, weighted F1 decreased by 0.0137. Validation-selected leave-class-out rejection recovered 0.403 unknown recall at 0.788 known coverage, varying by target. Cross-donor attributions were reproducible for most tested classes and enriched for canonical markers in 10/12 class–modality tests. On PDC101, MMoChi remained stronger.

Availability and Implementation: Stable public source and archive URLs will be supplied before submission.

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Supplementary information: Supplementary data will be available online.

Keywords single-cell annotation, CITE-seq, multimodal learning, donor shift, missing modality, interpretability

Abbreviations ACC: accuracy; ADT: antibody-derived tag; CITE-seq: cellular indexing of transcriptomes and epitopes by sequencing; HSR: hierarchy-aware sibling refinement; KD: knowledge distillation; M-F1: macro F1; W-F1: weighted F1

Introduction

Single-cell RNA sequencing and CITE-seq jointly measure transcriptomic programs and surface-protein markers at cellular resolution [33, 5, 37, 16, 36, 28]. RNA captures broad lineage and cell-state programs, whereas antibody-derived tags (ADTs) directly measure canonical surface markers. Integrated reference maps now resolve increasingly fine immune populations [41, 39], but supervised annotation becomes difficult when labels separate closely related naive, central-memory, effector-memory, regulatory and activated states. These boundaries are often biologically meaningful but weakly separated, and annotation errors can change downstream estimates of cell-state abundance, disease association and inferred immune programs.

Most annotation benchmarks still emphasize cell-level random splits. Such splits can place cells from the same donor in training and test, leading performance to combine annotation ability with donor-specific reuse. In practical reuse, the reference and target studies

may differ in donor composition, antibody panel, disease status and missing cell states. A model that performs well under a random split can therefore fail silently when the protein panel changes, when only RNA is available, or when a target population was absent from the reference. Aggregate accuracy also hides rare-class errors and gives little indication of which modality or feature supported a decision.

Reference classifiers, label-transfer methods and semi-supervised systems include CellTypist, scmap, scPred, scClassify and scNym [3, 19, 2, 23, 18, 9, 1]. TOSICA embeds prior knowledge in a transformer [7]; scVI, scANVI, totalVI and transfer-learning systems provide probabilistic representation learning and multimodal integration [25, 42, 12, 11, 26]. Benchmark rankings vary with task, modality and preprocessing, as do correction methods for batch and cohort shift [15, 20, 27, 4, 24]. In parallel, cell–cell communication tools in Bioinformatics commonly frame software contributions around explicit workflows, interpretable outputs, biological use cases and clearly delimited assumptions rather than a single headline

score. MMoChi uses a biologically specified multimodal classification hierarchy [6], and starCAT interprets T-cell programs through curated expression programs [21]. These approaches motivate evaluation that distinguishes predictive performance, robustness and biological interpretability.

We introduce MOSAIC, a compact dual-modal classifier designed to expose its prediction path. RNA, ADT and fused representations receive separate supervision; a training-only teacher supplies soft targets; branch logits are combined using top-two margins; and a bounded hierarchy-aware sibling-refinement (HSR) head can modify only prespecified local labels. Attention and sparse-expert models motivate modular prediction paths [40, 34, 43], but MOSAIC uses compact multilayer perceptrons to separate interface design from backbone scale. Unlike an unrestricted post-processing rule, every HSR action and its support can be enumerated.

The study asks four linked questions. First, does the model generalize to held-out donors under train-only preprocessing? Second, how does a frozen checkpoint degrade under protein-panel mismatch or RNA-only input? Third, can validation-selected rejection and parent fallback reduce leave-class-out failure? Fourth, are feature attributions stable across donors and enriched for prespecified markers? We also compare directly with official MMoChi on its compatible PDC101 sorted holdout. We report negative ablations and protocol boundaries alongside favorable results. The intended contribution is therefore an auditable multimodal annotation workflow evaluated under donor shift and stress tests, not a claim that every proposed module independently improves accuracy or that one model is universally state-of-the-art.

System and methods

Datasets and nested donor protocol

The primary dataset is the paired PBMC 3P cohort from GSE164378 [16]. It contains 161,764 cells from eight donors, 224 proteins and 58 fine-grained L3 labels. For each fold, 3,000 RNA genes were selected using training donors only. We used nested leave-one-donor-out evaluation. For test donor P1–P8, the validation donor was fixed as the next donor in a cycle (P1→P2, ..., P8→P1), and the remaining six donors formed training. Test sets ranged from 14,660 to 26,250 cells. All eight training folds contained all 58 labels, so the natural donor experiment tests donor shift under a closed-set label space.

Cells were partitioned before preprocessing. Gene selection, RNA transformations and RNA/ADT scalers were fitted using the six training donors and applied unchanged to validation and test. Model checkpoints, training hyperparameters and any rejection thresholds were selected without test labels. The eight held-out donors, not cells or model seeds, are the biological replicates for the primary statistical comparison. This design intentionally sacrifices the large nominal sample size of cell-level testing in exchange for a stricter estimate of across-donor generalization.

For a hierarchy-aware external comparator, we used the GSE229791 PDC101 sorted holdout with the official MMoChi hierarchy and `external_holdout=True` [6]. Hash-tag oligonucleotide (HTO), isotype and control proteins were excluded because PDC101 sort labels are linked to HTO/source information. MOSAIC used the identical 2,098-cell holdout and the same processed RNA/ADT input scale; scalers were fitted on training cells only. This comparison

is compatible at the holdout and label-hierarchy level, but it is not pooled with the PBMC donor analysis because the sorted PDC101 task has a different label space and uncertainty mode.

Model

For cell i , RNA and ADT inputs are $\mathbf{x}_i^{(r)}$ and $\mathbf{x}_i^{(p)}$. Modality encoders produce $\mathbf{h}_r = f_r(\mathbf{x}^{(r)})$ and $\mathbf{h}_p = f_p(\mathbf{x}^{(p)})$, and a fusion encoder maps their concatenation to $\mathbf{h}_f = f_f([\mathbf{h}_r; \mathbf{h}_p])$. Linear heads produce L3 logits $\mathbf{z}_r$, $\mathbf{z}_p$ and $\mathbf{z}_f$. Separate cross-entropy losses keep the modality branches predictive, so the model can report RNA-only, ADT-only and fused predictions from the same checkpoint. This branch design is used as an audit interface as much as a predictor: it exposes whether a cell is classified by a single dominant modality or by agreement between modalities.

The arbitration signal for branch b is its top-two logit margin $m_b = z_b^{(1)} - z_b^{(2)}$. Learned branch bias β_b and temperature τ define

$$w_b = \text{softmax}_b((m_b + \beta_b)/\tau), \quad \mathbf{z} = \sum_{b \in \{r, p, f\}} w_b \mathbf{z}_b. \quad (1)$$

This margin is an ordering signal, not a calibrated probability of branch correctness.

HSR receives the fused embedding and detached base probabilities. A delta head predicts local corrections only for two fixed sibling groups: CD4 TEM_3/CD4 TEM_4/CD4 TCM_3 and CD8 Naive/CD8 Naive_2/CD8 TCM_1. A bounded gate produces

$$g = 0.02 + 0.18 \sigma(\text{MLP}([\mathbf{h}_f; \text{stopgrad}(\text{softmax}(\mathbf{z}))])), \quad \mathbf{z}^{\text{final}} = \mathbf{z} + g \Delta \mathbf{z}. \quad (2)$$

All correction entries outside the six declared labels are exactly zero. HSR is an inspectable local intervention; it is not an open-set detector.

Training, ablations and statistical analysis

Each donor fold used seeds 41, 42 and 43. An early-fusion MLP was retrained for every seed and served as the matched baseline because it has the same RNA–ADT feature access without branch arbitration or HSR. The seed-42 MLP in each fold supplied stored training-cell probabilities to MOSAIC; validation and test teacher probabilities were never training inputs. We retrained four methods: MLP, MOSAIC full, MOSAIC without HSR and MOSAIC without KD. Full checkpoints were also evaluated without retraining as RNA branch, ADT branch, fusion branch, uniform fusion, learned margin gate and learned margin gate plus HSR. The retrained ablations test whether training-time modules improve a deployable model, whereas the same-checkpoint ablations isolate inference-time arbitration and HSR effects without changing learned representations.

The implemented objective combines sqrt-balanced cross-entropy, branch supervision, temperature-scaled KL distillation and the local HSR loss:

$$\mathcal{L} = \mathcal{L}_{\text{wCE}} + \lambda_b(\mathcal{L}_r + \mathcal{L}_p) + \lambda_f \mathcal{L}_f + \lambda_{\text{KD}} T^2 D_{\text{KL}}(\mathbf{q} \parallel \mathbf{p}^T) + \lambda_{\text{HSR}} \mathcal{L}_{\text{HSR}}. \quad (3)$$

Models used AdamW for at most 35 epochs, batch size 1,024, learning rate 10^{-3} , weight decay 10^{-4} , dropout 0.2 and patience 14. We set $\lambda_b = 0.35$, $\lambda_f = 0.50$, $\lambda_{\text{KD}} = 0.05$, $T = 2$ and $\lambda_{\text{HSR}} = 0.20$. Checkpoints maximized validation M-F1 subject to 0.0015 tolerances for validation ACC and W-F1.

MOSAIC architecture for hierarchy-aware multimodal single-cell annotation

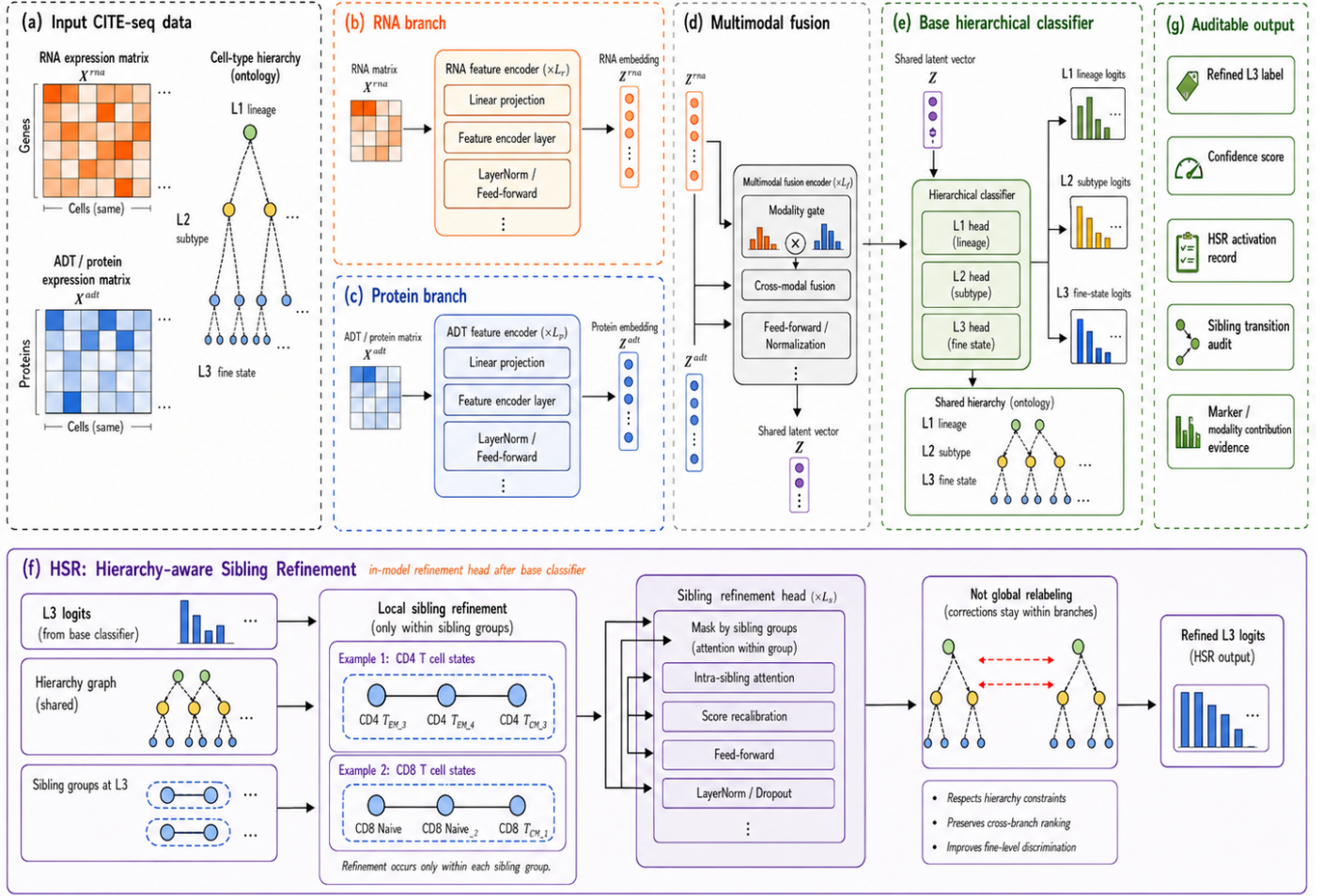


Figure 1 MOSAIC architecture. (a) Paired CITE-seq RNA and ADT matrices and the cell-type hierarchy. (b,c) RNA and protein branches encode modality-specific representations. (d) Multimodal fusion combines both embeddings into a shared representation. (e) The base hierarchical classifier produces lineage, subtype and fine-state outputs. (f) Hierarchy-aware sibling refinement restricts local changes to prespecified sibling groups before producing refined L3 logits. (g) Auditable outputs summarize the refined label, confidence, HSR activity, sibling transitions and modality or marker evidence. **Alt text:** A multimodal single-cell architecture diagram shows paired RNA and protein inputs, separate encoders, multimodal fusion, hierarchical classification, local sibling refinement and auditable outputs.

We report ACC, W-F1, M-F1, balanced accuracy and per-class F1. Seeds were averaged within each held-out donor; donor means, 95% Student-*t* intervals, paired-difference intervals and Wilcoxon sensitivity tests were then calculated across eight donors. No cell-level bootstrap was used for the primary PBMC comparison because cell bootstrap intervals would treat thousands of cells from the same donor as if they were independent biological replicates. We interpret the donor-paired interval, rather than the largest point estimate in a table, as the primary evidence for a method difference.

Incomplete panels, rejection and attribution

Each of the 24 full checkpoints was frozen and evaluated with full input; deterministic feature-level masks removing 10, 30, 50 or 70% of proteins; prespecified memory-marker and T-cell-marker groups; and RNA-only input. Masks and random seeds were fixed before test evaluation. We measured performance, expected calibration

error (ECE), multiclass Brier score and degradation slope. These experiments are stress tests of a full-input model under panel mismatch, not mask-aware training experiments. They are included because antibody panels are often study-specific and because a useful reference classifier should fail gradually and transparently when ADT information is reduced.

Because every natural donor fold retained all 58 labels in training, unknown-label evidence used five prespecified leave-class-out targets: gdT.2, NK.3, CD4 TCM.1, CD8 TEM.4 and B naive lambda. Each model was retrained on 57 labels. Thresholds for one minus maximum probability, one minus top-two margin and energy were selected using known validation cells to target 0.95 or 0.80 known coverage. These scores are related to confidence calibration, uncertainty estimation and selective prediction [14, 22, 10, 17, 13]. We report unknown recall, AUROC/AUPRC, accepted L3 accuracy, parent fallback and

combined hierarchical accuracy. This is a pseudo-unknown stress test rather than unrestricted open-world evaluation.

Gradient-times-input attribution was computed for up to 64 correctly predicted cells per donor, seed and sibling label. RNA and ADT features were summarized separately. We quantified donor-pair rank Spearman correlation and top-20 Jaccard overlap after averaging seeds. Canonical-marker top-20 enrichment used prespecified class-modality marker sets and 5,000 permutations. Integrated gradients on a small seed-42 subset checked directional agreement. These analyses complement established local-surrogate, integrated-gradient, relevance-propagation and Shapley approaches [31, 38, 30, 32, 29]; they quantify computational stability and marker concordance, not causal regulation.

Implementation

The Python/PyTorch implementation saves configuration, split manifest, selected checkpoint, predictions, per-class metrics, run log and SHA-256 artifact index for every training unit. Table and figure scripts consume V33 CSV/JSON artifacts only and reject incomplete matrices. This artifact contract is part of the method because the paper reports several negative and stress-test results; every table row must therefore be traceable to a declared run rather than manually copied from logs.

The manuscript-facing release candidate contains the exact table fragments, figure source manifests, split definitions and audit scripts used to generate the main text and supplement. Raw data, caches and checkpoints are excluded from the local release package, but their source accessions and checksum-indexed derived artifacts are recorded. Experiments used one NVIDIA RTX 4090. A software license, test data, stable repository URL and archived release remain required before submission.

Results

Nested donor-disjoint annotation

The first analysis asks whether the proposed workflow improves annotation when the test donor is absent from all preprocessing and training steps. Across eight held-out donors, MOSAIC full reached ACC 0.9119 (95% CI 0.9086–0.9153), W-F1 0.9108 (0.9062–0.9153) and M-F1 0.8072 (0.7917–0.8227; Table 1; Fig. 2a). Relative to matched MLP, paired donor differences were +0.0029 ACC (CI –0.0001 to 0.0059), +0.0040 W-F1 (0.0013–0.0066) and +0.0148 M-F1 (0.0091–0.0206). Seven of eight donors improved in W-F1, and all eight improved in M-F1. The ACC interval crossed zero and is not claimed as significant.

This pattern is consistent with a rare-class and class-balance benefit rather than a broad increase in every cell-level decision. Weighted F1 improved in most donors, macro F1 improved in every donor and balanced accuracy improved relative to MLP, but the largest classes dominate ACC enough that the ACC interval remains inconclusive. We therefore use the donor-paired W-F1 and M-F1 results as the main positive performance evidence, and avoid ranking methods by unpaired point estimates alone.

Direct module attribution is negative

Direct ablations did not identify HSR, KD or learned margin arbitration as sources of the aggregate improvement over MLP.

Retrained no-HSR exceeded full by 0.0016 ACC and 0.0025 M-F1; no-KD exceeded full by 0.0015 ACC and 0.0038 M-F1. Within the same full checkpoints, uniform fusion exceeded learned margin gate plus HSR on all eight donors by 0.0011 ACC, 0.0011 W-F1 and 0.0042 M-F1 (Fig. 2b). Enabling HSR within the same checkpoint changed aggregate metrics by less than 0.0005 and produced mixed directions.

These negative results are important for interpreting the architecture. The branch-supervised model is useful as an auditable multimodal classifier, but this experiment does not prove that the bounded HSR head or learned gate is the mechanism behind the donor-level gain. HSR and margin gating provide explicit, enumerable behavior: the model can state whether a local sibling correction was available, whether it fired and which labels were affected. That inspectability is retained as a software and workflow contribution, while the current aggregate performance claim is attributed to the overall branch-supervised training setup rather than to a single positive module.

Incomplete proteins produce graceful but nontrivial degradation

At 50% random protein masking, W-F1 was 0.8971, a donor-level decrease of 0.0137 from full input; at 70% masking it was 0.8921 (Fig. 2c). RNA-only input reached W-F1 0.8882, a decrease of 0.0226. M-F1 was more sensitive, decreasing by 0.0330 at 50% masking and 0.0514 for RNA-only input. ECE increased from 0.0713 to 0.1063 for RNA-only input.

The result shows gradual degradation rather than abrupt collapse. This is useful in practice because CITE-seq references are often reused with incomplete or mismatched protein panels. However, rare labels and calibration are more fragile than W-F1, and marker-group masks can be more damaging than random masks at the same feature count. These frozen-checkpoint tests therefore support a limited robustness statement: MOSAIC remains operational and measurable under missing proteins, but it has not been trained to be invariant to arbitrary panel loss.

Leave-class-out rejection is target dependent

At target known coverage 0.80, one-minus-margin rejection achieved observed known coverage 0.7878, mean unknown recall 0.4032 and combined hierarchical accuracy 0.9146 (Fig. 2d). Energy gave similar recall (0.4024) but lower combined hierarchical accuracy (0.9073). Results varied sharply by target: mean margin-based recall was approximately 0.57 for CD4 TCM.1 and gdT.2 but 0.13 for B naive lambda. Unknown parent-safe rate was 0.2259 overall.

The leave-class-out experiment addresses a different failure mode from donor shift. The model is forced to classify cells whose fine label was removed from the reference, and the rejection rule is selected only from known validation cells. The target-dependent outcome indicates that confidence scores contain useful information for some absent labels but not enough for a general open-world claim. Rejection therefore exposes a coverage-risk option and a hierarchy fallback interface, but it does not solve open-set recognition.

Compatible hierarchy baseline and quantitative attribution

On the locked PDC101 holdout, MOSAIC achieved 0.9077 ACC, 0.9068 W-F1 and 0.9007 M-F1, while official MMoChI achieved 0.9347, 0.9335 and 0.9298 (Fig. 2e). The uncertainty modes differ:

Table 1 Nested donor-disjoint performance. Values are held-out-donor means $\pm$ 95% Student-*t* margins after averaging three seeds within each donor. The highest point estimate is not interpreted as a significant ranking without a paired interval.

Metric	MLP early fusion	MOSAIC without KD	MOSAIC without HSR	MOSAIC full
accuracy	0.9091 $\pm$ 0.0043	0.9134 $\pm$ 0.0033	0.9135 $\pm$ 0.0033	0.9119 $\pm$ 0.0033
weighted_f1	0.9068 $\pm$ 0.0048	0.9121 $\pm$ 0.0043	0.9121 $\pm$ 0.0042	0.9108 $\pm$ 0.0045
macro_f1	0.7923 $\pm$ 0.0165	0.8110 $\pm$ 0.0154	0.8097 $\pm$ 0.0146	0.8072 $\pm$ 0.0155

MOSAIC uses three-seed Student-*t* margins, whereas MMoChi uses a cell bootstrap conditional on one workflow fit. The result establishes a compatible direct comparison and a negative method boundary. It also clarifies the task distinction: a curated hierarchy with marker-aware rules can be stronger in a small sorted label space, whereas MOSAIC is designed for high-dimensional reference-trained annotation and auditability under donor and modality stress tests.

Across donor pairs, mean attribution Spearman correlation ranged from 0.529 to 0.796 for RNA and from 0.295 to 0.612 for ADT (Fig. 2f). Ten of 12 class-modality tests showed canonical-marker enrichment at unadjusted permutation $P < 0.05$. CD4 TEM4 ADT and CD8 Naive RNA did not. Seed-level attribution stability and integrated-gradient checks were generally stronger than donor-pair overlap, indicating that donor variation, not only model initialization, limits interpretation. The attribution analysis is therefore quantitative evidence of reproducibility and marker concordance for selected sibling classes, not a discovery claim for new biomarkers.

Discussion

The primary result is a donor-disjoint, leakage-controlled evaluation of fine-grained multimodal annotation. MOSAIC improved W-F1 and M-F1 over a matched early-fusion MLP across donors, while the ACC interval crossed zero. This distinction matters for fine immune annotation: rare and sibling labels contribute little to overall accuracy but strongly affect macro-level and balanced metrics. Protein masking showed that a full-input model can degrade gradually, but rare-class performance and calibration remain sensitive when the ADT panel disappears. Leave-class-out rejection was useful for some targets and weak for others.

The architectural contribution should be interpreted narrowly. Branch supervision exposes modality-specific competence, and HSR restricts any local correction to declared sibling labels. This makes intervention support auditable: a user can inspect branch logits, gate behavior, affected sibling groups and whether a local correction changed the final label. However, no-HSR, no-KD and uniform fusion performed as well as or better than the full configuration. It would therefore be incorrect to attribute the MLP comparison gain to HSR, distillation or learned gating. The honest conclusion is that the current architecture provides a structured audit interface and competitive donor-level macro performance, while the individual HSR and gating modules require redesign before being claimed as performance-improving components.

MMoChi remained superior on its native sorted hierarchy. This is consistent with a marker-defined method being effective

in a small, structured label space. MOSAIC addresses high-dimensional reference classification, but that distinction does not erase the negative comparison. Similarly, canonical-marker enrichment supports biological concordance for selected labels, not discovery of causal biomarkers. These results place MOSAIC between two extremes: it is more inspectable than a monolithic early-fusion classifier, but it is not a substitute for a carefully curated marker hierarchy when that hierarchy matches the task.

Limitations include only eight PBMC donors, one compatible direct hierarchy baseline, complete training-label support in natural donor folds, and post hoc masking rather than mask-aware training. Leave-class-out targets are constructed pseudo-unknowns, PDC101 uncertainty modes are not interchangeable, and donor-pair attribution statistics are descriptive because pairs overlap. The older random-cell, patient-disjoint COVID-19 [35], 5P assay-cohort and COMBAT analyses are retained in the Supplement as secondary breadth evidence and are not pooled with the primary donor-level inference.

Future work should focus on the modules that were not validated here. A panel-aware training objective could replace post hoc masking and explicitly teach RNA-only or partial-ADT fallback. HSR should be redesigned with validation-selected activation policies or hierarchy-aware loss terms that prove benefit before test evaluation. Unknown handling should move from leave-class-out stress tests to prospective external datasets with true absent or novel states. Finally, attribution evidence should be extended from stability and marker overlap to perturbation-based checks that test whether identified features affect model decisions in predictable directions.

Conclusion

MOSAIC provides a reproducible framework for fine-grained RNA-ADT annotation with explicit modality branches, bounded local intervention and cell-level audit outputs. Nested donor evaluation supports W-F1 and M-F1 gains over matched early fusion, while direct ablations delimit rather than validate the proposed HSR, KD and gating mechanisms. Missing-panel, pseudo-unknown, PDC101 and attribution analyses define where the current model is reliable and where additional methodological work is required.

Data availability

PBMC 3P and 5P data are from GEO accession GSE164378. The COVID-19 cohort is from ArrayExpress E-MTAB-10026. PDC101 is from GEO GSE229791. COMBAT is available through Human Cell Atlas project cdabcf0b-7602-4abf-9afb-3b410e545703, EGA study EGAS00001005493 and Zenodo DOI 10.5281/zenodo.6120249

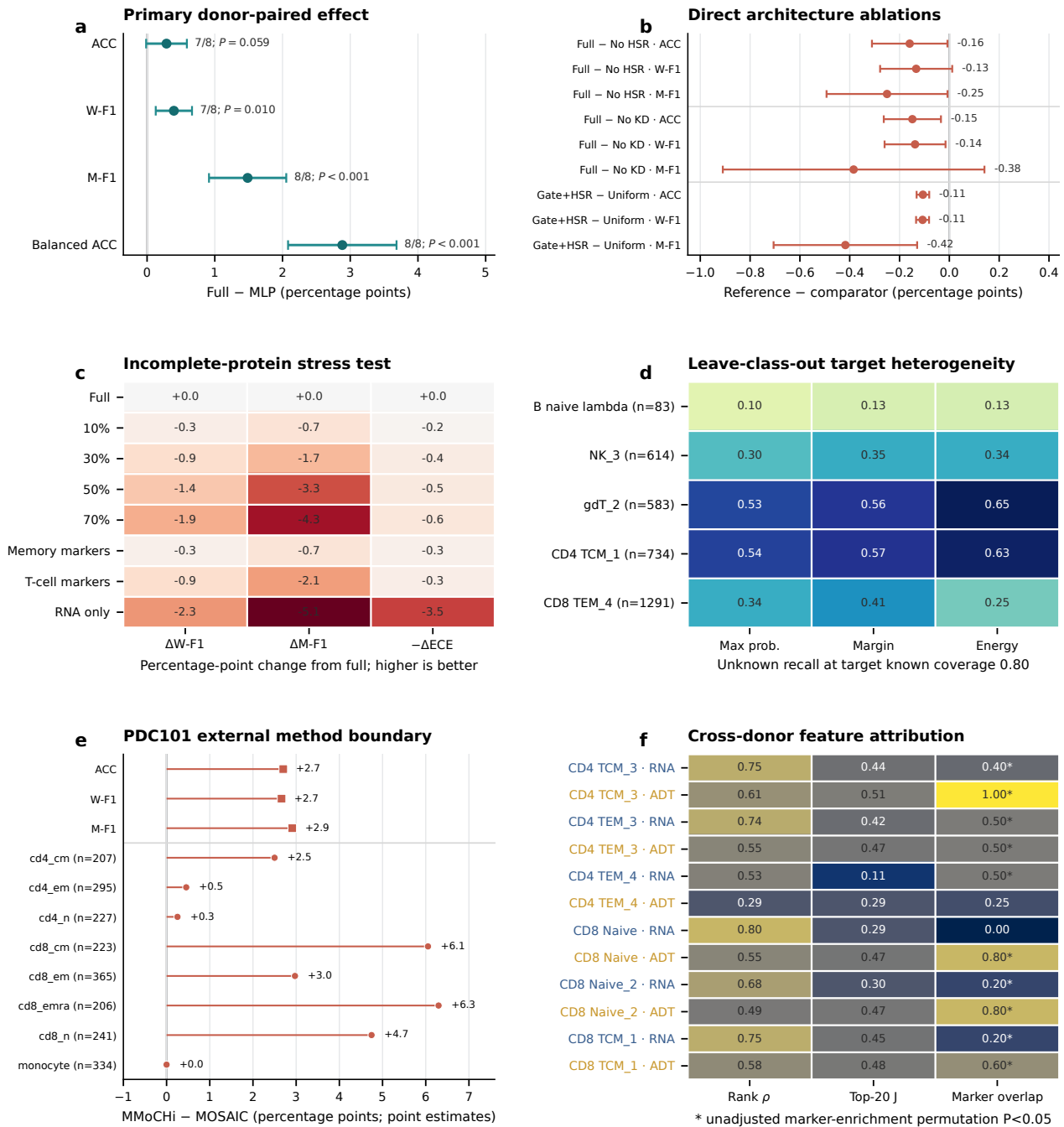


Figure 2 Donor-disjoint performance, direct ablations and evidence boundaries. (a) Donor-paired MOSAIC full minus MLP differences with 95% intervals, positive-donor counts and paired P values. (b) Direct retrained and same-checkpoint module comparisons; negative values favor the comparator. (c) Percentage-point changes in W-F1, M-F1 and sign-reversed ECE relative to full protein input under deterministic panel masks. (d) Unknown recall for five leave-class-out targets and three validation-selected reject scores at target known coverage 0.80. (e) Aggregate and per-class PDC101 point-estimate differences between official MMoChi and MOSAIC; uncertainty modes are not pooled. (f) Donor-pair attribution rank stability, top-20 overlap and canonical-marker overlap for six sibling classes and both modalities. **Alt text:** Six panels show positive macro-level donor-paired effects over MLP, negative or null direct module ablations, gradual degradation under protein removal, target-dependent unknown recall, lower PDC101 scores than MMoChi and heterogeneous cross-donor feature-attribution stability.

[8]. Exact processed split manifests and source-record checks are included in the local release candidate; public archive identifiers must be inserted before submission.

Code availability

The source code, configuration files, test data and documentation required by the Bioinformatics software policy must be released at a stable public URL before submission. The local V33 release candidate

includes commands, environment metadata, split manifests, evidence tables, tests and checksums without raw data or checkpoints.

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Author contributions

CRediT roles must be confirmed by all authors before submission.

Conflict of interest

The authors declare no competing interests.

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OpenAI Codex was used during internal preparation for code assistance, artifact auditing, table generation and language-draft assistance. The human authors retain responsibility for the analyses, citations, figures and conclusions. Under the current Bioinformatics policy, this working draft must be independently reviewed and rewritten by the authors before submission, and the final permitted use of AI-assisted tools must be disclosed in both the manuscript and cover letter.

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